A Geometric Framework for Understanding Memorization in Generative Models

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Background

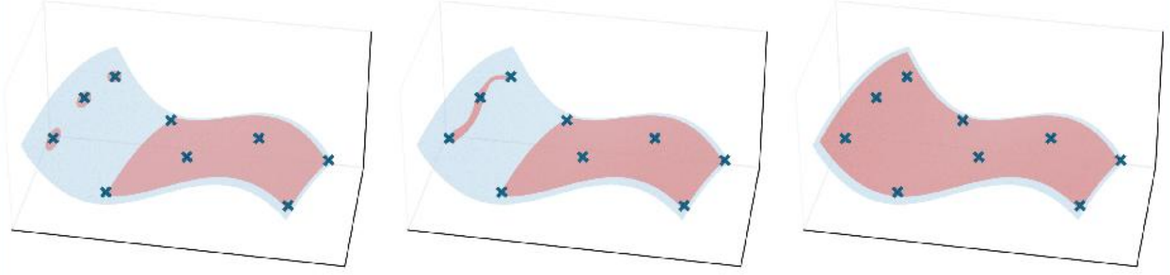
Memorization and Overfitting in Deep Generative Models

- Deep generative models are known to memorize and reproduce their training data
- This behaviour reduces model utility and may have severe privacy and legal implications
- Contributing factors [4]: data complexity, duplication of training points, and highly specific conditioning
- There is as yet no systematic framework describing memorization in DGMs

The Manifold Hypothesis

- High-dimensional data lies on a manifold $\mathcal{M}_*$ with multiple components of varying dimensionalities
- DGMs also generate samples from a manifold $\mathcal{M}_\theta$ with the goal of matching $\mathcal{M}_\theta$ to $\mathcal{M}_*$
- The local intrinsic dimension of $x \in \mathcal{M}_*$ (resp. $x \in \mathcal{M}_\theta$), denoted $\text{LID}_*(x)$ (resp. $\text{LID}_\theta(x)$), is the dimension of the manifold component containing x
- Intuitively, the LID of x captures the complexity of the datapoint x and constraints on the dataset's structure

Our Geometric Framework



(a) Exact memorization ($\text{LID}_\theta(x) = 0$) (b) Some Memorization ($\text{LID}_\theta(x) = 1$) (c) No memorization ($\text{LID}_\theta(x) = 2$)

Figure 1: LID values for three red model manifolds with different degrees of memorization of the three leftmost datapoints, which are depicted as dark blue crosses. The true manifold is light blue.

We propose that $x \in \mathcal{M}_*$ is memorized when $\text{LID}_\theta(x)$ is small.

- This encompasses both exact copying of training images ($\text{LID}_\theta(x) = 0$) and copying of styles or parts of an image as in Figure 2 ($\text{LID}_\theta(x) > 0$ but still small)

From this definition, memorization can be split into two types:

1. $\text{LID}_\theta(x) < \text{LID}_*(x)$: *overfitting-driven memorization* (OD-Mem), in which the model fails to generalize to the ground truth manifold correctly (Figure 1)
2. $\text{LID}_\theta(x) = \text{LID}_*(x)$: *data-driven memorization* (DD-Mem), in which the ground truth data distribution has been learned correctly, but itself contains very few degrees of freedom at x

Figure 2: 8 images along a relatively low-dimensional manifold learned by Stable Diffusion v1.5. The first is a real image from LAION (flagged as memorized by [2]), and the remainder were generated by the model.

Explaining Memorization

The geometric framework explains memorization-related phenomena in the literature:

Duplicated Data It has been observed that memorization occurs when training points are duplicated. Here we identify that data duplication is equivalent to an LID of 0:

Theorem (Informal). With high probability and a sufficiently large dataset, duplicate samples occur at x_0 if and only if $\text{LID}_*(x_0) = 0$.

Complexity. [4] observes that simple images are memorized more often. This is neatly explained by the fact that they have lower LID_*.

Conditioning. Highly-specific conditioning inputs encourage the reproduction of memorized samples. We explain this by the fact that conditioning on a prompt c decreases LID:

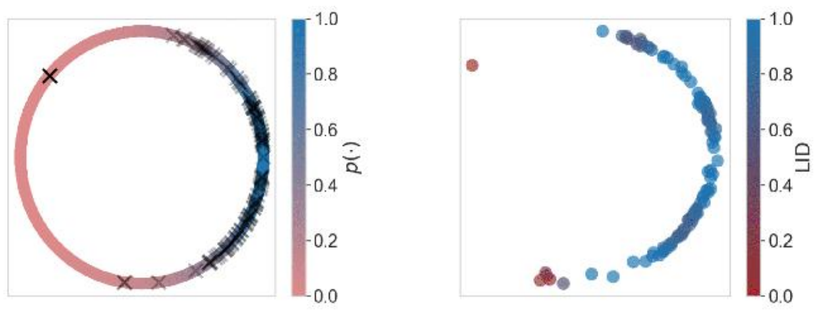
Proposition. $\text{LID}_\theta(x|c) \leq \text{LID}_\theta(x)$.

Classifier-Free Guidance (CFG). A large classifier-free guidance vector in diffusion models correlates strongly with memorization [5]. This is explainable by observations in the literature that faster explosion of model norms during sampling corresponds to lower LID_θ.

Empirical Verification

We can test this framework using the LID_θ estimator of [3]:

von Mises Distribution We train a diffusion model on a von Mises distribution; in Figure 3b we depict model samples with their corresponding LIDs, and a couple of samples are clearly memorized with $\text{LID}_\theta(x) = 0$, while the remaining samples have $\text{LID}_\theta(x) = 1$.



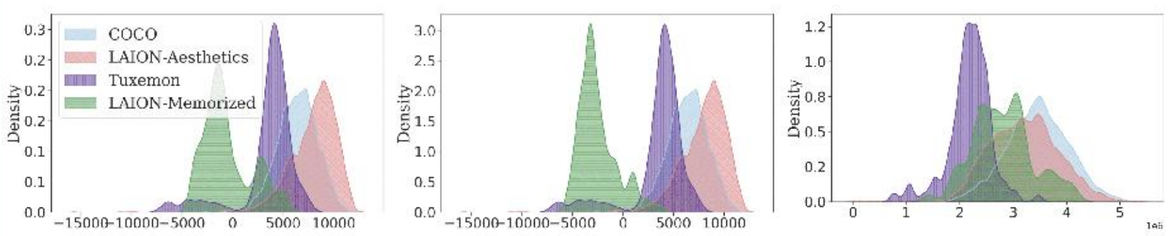
(a) Ground truth manifold. (b) Model samples.

Figure: Training a diffusion model on a von Mises distribution.

Stable Diffusion on LAION, COCO, and Tuxemon We compute LID_θ of Stable Diffusion [1] on

- Unmemorized images from a mix of LAION Aesthetics 6.5+, COCO, and Tuxemon
- Memorized images from LAION [2]

We find that low LIDs correspond better to memorized images than PNG size, a proxy for image complexity.



(a) Unconditional LID (AUROC = 97.6%). (b) Conditional LID (AUROC = 99.2%). (c) PNG compression (AUROC = 70.3%).

Figure 4: Density histograms for each memorization metric on the different datasets.

[1] Rombach, Blattmann, Lorenz, Esser, Ommer. "High-resolution image synthesis with latent diffusion models". CVPR 2022.

[2] Webster. A reproducible extraction of training images from diffusion models. arXiv 2305.08694.

[3] Kamkari, Ross, Hosseinzadeh, Cresswell, Loaiza-Ganem. "A geometric view of data complexity: Efficient local intrinsic dimension estimation with diffusion models". arXiv 2406.03537.

[4] Somepalli, Singla, Goldblum, Geiping, Goldstein "Understanding and mitigating copying in diffusion models". NeurIPS 2023.

[5] Wen, Liu, Chen, Lyu "Detecting, explaining, and mitigating memorization in diffusion models". ICML 2023.